# Machine Learning

**What is machine learning ?**
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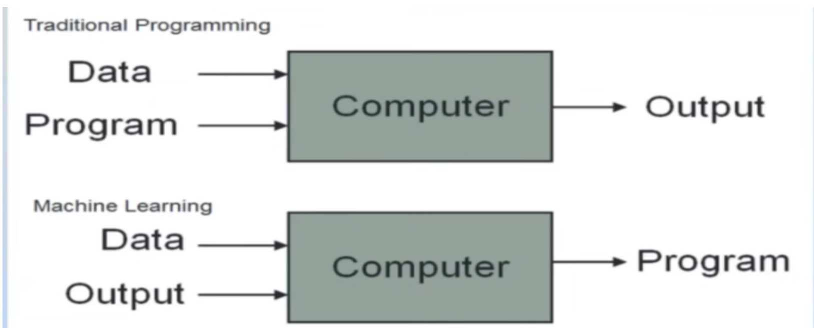
Machine learning is a field of computer science that uses
statistical techniques to give computer systems the ability to "learn" with data, without being explicitly programmed
Machine Learning is the process of teaching computers to learn from data and make predictions or decisions without explicit programming.

**Flow chart**




**Where we use machine learning**

Machine learnig is used in many industries to predections, automate tasks, and improve decision-making.

- **Healthcare** – Disease prediction, medical image analysis
- **Banking & Finance** – Fraud detection, credit score prediction
- **E-commerce** – Product recommendations (Amazon, Flipkart)
- **Social Media** – Friend suggestions, content recommendations
- **Transportation** – Self-driving cars, traffic prediction

**Life cycle of machine learning**

1. **Problem Definition**
    a. Identify the business problem.
2. **Data Collection**
    a. Gather data from different sources.
3. **Data Preprocessing**
    a. Clean missing values, remove duplicates, and prepare the data.
4. **Exploratory Data Analysis (EDA)**
    a. Analyze patterns, trends, and relationships in the data.
5. **Feature Engineering**
    a. Select or create important features for the model.
6. **Model Selection**

    a. Choose an appropriate ML algorithm.
7. **Model Training**
    a. Train the model using training data.
8. **Model Evaluation**
    a. Test the model using metrics like Accuracy, Precision, Recall, and F1-score.
9. **Model Deployment**
    a. Deploy the model into a real-world application.
10. **Monitoring & Maintenance**

- Continuously monitor and update the model as new data becomes available.

**What is data analysis**

Data analysis is the process of collecting, cleaning, organizing, and interpreting data to discover useful information and support decision-making.

**Why machine learning important**

Machine Learning is important because it enables computers to learn from data without being explicitly programmed. Like,

- Makes accurate predictions
- Handles large amounts of data
- Improves decision-making
- Detects fraud and cyber threats

**History of machine learning**

- **1950** – Alan Turing proposed the **Turing Test** to evaluate machine intelligence.
- **1952** – Arthur Samuel developed a checkers program that learned from experience.
- **1957** – Frank Rosenblatt introduced the **Perceptron**, an early neural network.
- **1960s–1970s** – Research focused on pattern recognition and early AI.
- **1986** – Backpropagation made training neural networks more effective.
- **1990s** – Algorithms like Decision Trees, Support Vector Machines (SVM), and Random Forest gained popularity.
- **2000s** – Big data and faster computers accelerated ML development.
- **2010s** – Deep Learning became successful in image recognition, speech recognition, and natural language processing.
- **Today** – Machine Learning powers AI applications such as ChatGPT, self-driving cars, recommendation systems, healthcare, and financial analytics.

**How machine learning works**

- Collect data.
- Clean and prepare the data.
- Train a machine learning model.
- Test the model.
- Use the trained model to make predictions on new data.